

WASP-43b

R-0384

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_250324 · created 2026-07-18T20:52:52+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460758.6961 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.015
Transit depth (Rp/Rs)^2	1.88 +/- 0.41 [%]
Orbital Inclination (inc)	85.54 +/- 1.81
Airmass coefficient 1 (a1)	0.9393 +/- 0.0099
Airmass coefficient 2 (a2)	0.04 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.06 %
Best Comparison Star	#3 - [98, 258]
Optimal Aperture	3.72
Optimal Annulus	9.57
Transit Duration (day)	0.0567 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.137 $\pm$ 0.015 vs 0.1594 (deviation 1.49 σ)
Duration fit vs published	0.0567 d vs 0.0512 d (deviation 1.82 σ)
Mid-time O–C	3.8 min
Depth SNR	9.1
Residual scatter	1.06 %

Light curve

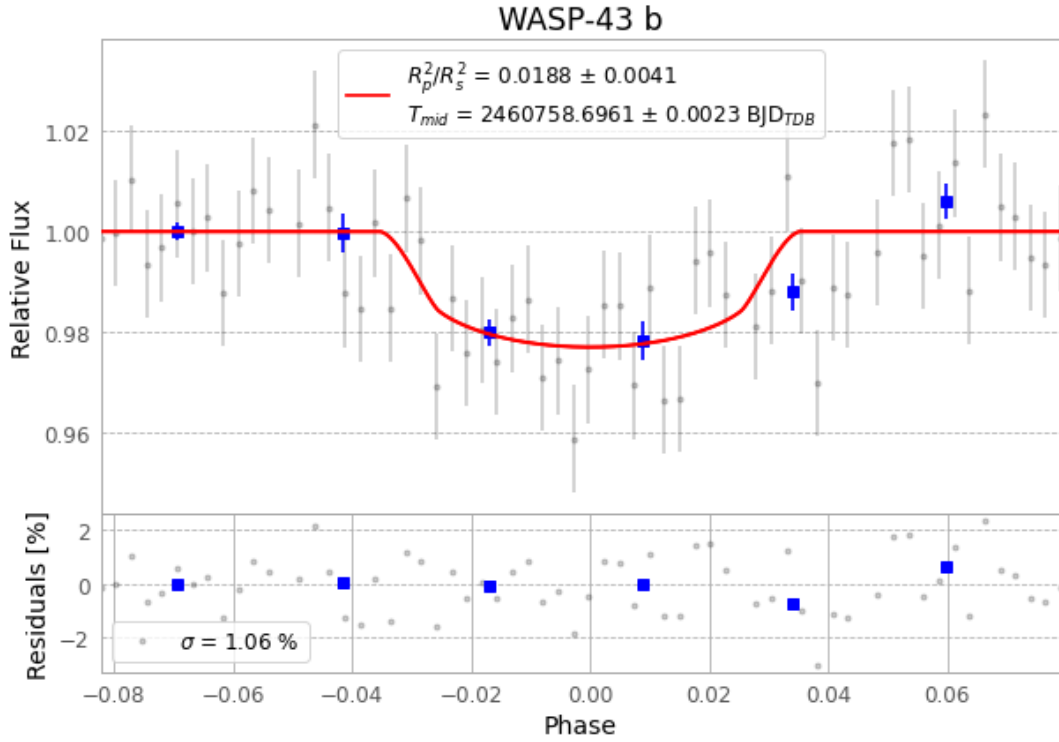


Figure 1 — FinalLightCurve_WASP-43 b_2025-03-23.png (SHA-256 a4d678cdb0432fa6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [400, 193], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1f3f3f45b6bc5a71...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

64 FITS frames (42 MB), WASP-43250324025716.FITS → WASP-43250324060616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250324.log (51518dec3460...), FinalLightCurve_WASP-43 b_2025-03-23.png (a4d678cdb043...), FinalLightCurve_WASP-43 b_2025-03-23.csv (0c3692ab0f80...)

Compute resources

Wall time 100.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 20:52Z.